

GUEST LECTURE · CYBR 1100

Don't Pick a Lane Just Yet

Planning your **cybersecurity career** in the age of AI agents



Adam · Pixee AI

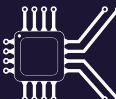
September 9, 2026



SECTION 0

Why listen to me



- 01 I build automated ai-powered code and security remediation at  Pixee.
- 02 I make automations and machines that do work that was a full job three years ago.
- 03 I hire, and I read the résumés that don't make it.
- 04 I've worked at the world's biggest scale at  to tiny 20-person startups.
- 05 3x International speaker, co-founder of DEF CON 402 and  Kernelcon

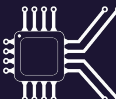
This talk wants arguments back. Bring them.

SECTION 0

Why listen to me

I have been wrong before. Publicly. Multiple times.



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Something Significant Happened Yesterday

This **blew my mind** and made me wonder if we are at an **AI inflection point**.

September 8, 2026: Millennium Prize

WHAT THE CLAIM SAYS

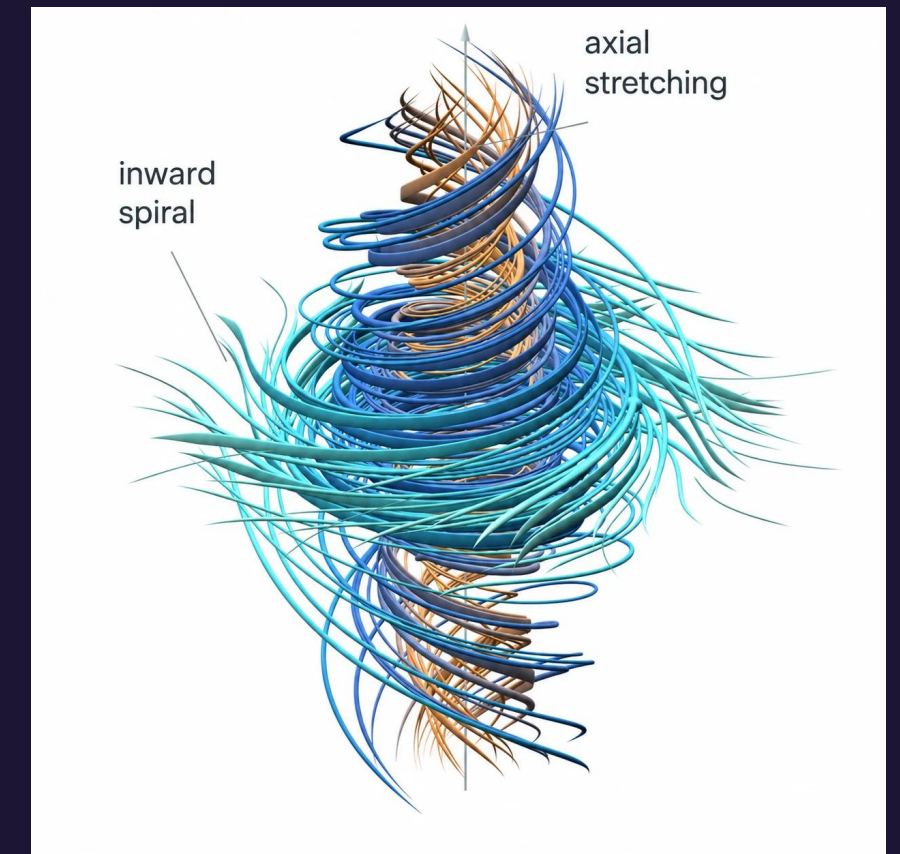
A smooth 3D fluid at rest, under smooth forcing with finite energy, can go singular in finite time. Formalized in Lean.

WHAT IT DOES NOT SAY

Smoothness breaks. That is not the same as predicting turbulence.

~10,000 coordinating agents ~88 hours unreleased internal model

<https://openai.com/index/navier-stokes-solution/>



The credit dispute

Aug 15, 2026

Buckmaster (NYU) and Alpöge prove Euler blow-up, using Anthropic models.

Sept 8, 2026

OpenAI's weekend push lands. Buckmaster alleges rumors of the unpublished work reached them first.

Who did the work?

How do we verify it?

What happens to unpublished work inside someone else's product?

THE HINGE

10,000 agents. 88 hours. One campaign.

The story wasn't that a model got smart. It ran a coordinated multi-day operation.

What else is a coordinated multi-day campaign?

Think about cybersecurity

What is Next?

The **future of AI** and possible technology careers (*according to me*).

Three forces

01

**The model moves
inside the perimeter**

02

**Code gets cheap,
judgment gets
expensive**

03

**Offense and defense
both run at machine
speed**

Two are documented. One is my prediction. I'll flag which.

The model moves inside the perimeter

WHY IT HAPPENS

Regulated data (HIPAA, CMMC, GDPR, ITAR). Sovereignty. Cost at volume. Air-gapped sites where a cloud API isn't an option.

Caveat: "on-prem" that phones home for licensing isn't air-gapped. Verification matters.

THE NEW ASSET CLASSES

GPUs

weights

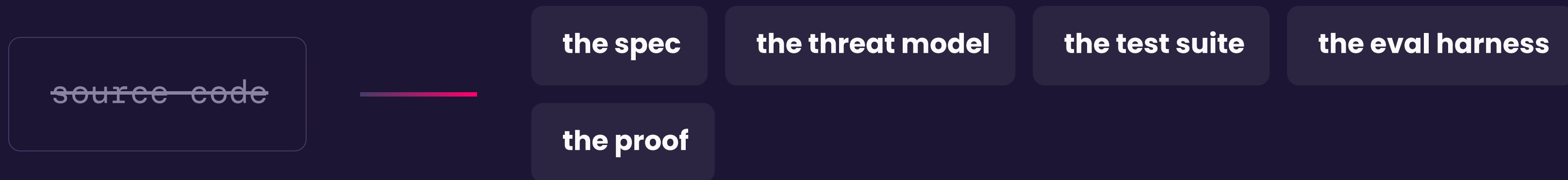
inference gateway

vector store

audit log

Somebody has to secure that box. It isn't a web app, and there's no mature playbook.

Code becomes ephemeral



Yesterday's Lean formalization is the example: a machine-checkable artifact let people trust output no human fully read.

Premium work: deciding what to build, deciding what "correct and safe" means, fixing what breaks under pressure.

FORCE 03

DOCUMENTED

Machine speed, both directions

~20

zero-days autonomously surfaced in open source by Google's Big Sleep, Nov 2024 → Aug 2025 (human-reviewed before reporting)

#1

HackerOne leaderboard position held by XBOW, an autonomous offensive security company, by mid-2025

DARPA AI CYBER CHALLENGE FINALISTS ·
~54M LINES ANALYZED

 **86%**
synthetic vulns found

 **68%**
patched · every finalist open-sourced

NIST's draft Cyber AI Profile describes agents autonomously running attack phases: recon, exploitation, credential harvesting, lateral movement.

**When standards bodies write it down,
hiring criteria change.**

Google Project Zero / DeepMind Big Sleep, 2024-25 · XBOW & HackerOne, 2025 · DARPA AIxCC finals, 2025 · NIST Cyber AI Profile (draft)

Where I might be wrong

Jevons paradox

Cheaper code means more code, more systems, more attack surface - and more security work, not less.

Business logic

The weakest spot for AI tools is the bug class that requires knowing what the application is for.

Legacy reality

Hospitals, utilities, manufacturing, avionics, firmware: 20–40 year old systems no model is rewriting. Somebody still has to understand Modbus.

Bad track record

Predictions of this shape age badly. COBOL was supposed to be dead. 4GLs were supposed to end programming.

These are the strongest arguments against that I know. If you have a better one, say it in Q&A.

Both can be true



The floor rises



The ceiling rises faster



**The middle gets
squeezed**

Adaptability is the hedge that pays off under either forecast. That's why it's the recommendation.

What is the same?

The **future of AI** that doesn't change (*according to me*).

What does not change

Networking and protocols

OS and memory internals

Cryptography and key management

Identity and authorization

Data modeling

Clear writing

Every AI-era problem on the next slides is one of these wearing a new hat.

Run models locally

start here	Ollama · LM Studio
tradeoffs	llama.cpp · quantization
serving	vLLM · OpenAI-compatible gateway
weights	Llama · Qwen · Mistral · gpt-oss
context + tools	RAG with a vector store · MCP servers
the part people skip	a written eval

Secure this stack and you graduate into a job nobody has enough people for.

Jobs that barely existed three years ago

AI red-teamer / eval engineer

Break the model, then measure it repeatably.

Agent & non-human identity

Credentials, scoping, revocation. The sleeper issue.

Prompt-injection & tool-abuse defense

The new injection class. Still unsolved.

AI supply chain & model provenance

Weights, AIBOMs, poisoned artifacts.

Detection engineering for agents

What does a rogue agent look like in the logs?

AI governance & assurance

For people who like policy more than packets.

Titles vary by employer. The work is real.

SAY IT OUT LOUD

The entry-level squeeze

Junior work is the most automatable work.
Pretending otherwise is dishonest.

Unverified as a labor-market claim · this is my read, not
a study

WHAT MAKES A JUNIOR VALUABLE NOW

Verification. Taste. Explaining a finding to someone who has to act on it.

A homelab you can defend the design of

- CTFs
- Real open-source contributions
- Writeups of what you did
- Being the person who says "this output is wrong, here's why"

PERSONAL EXAMPLE

My example

How my homelab helps me learn.



Okay, now what?

How can you prepare yourself for a future career in the AI space?

Two four-year paths

STUDENT A

Pentest labs. Only pentest labs.

Competes with a tool that improves every quarter, on the tool's own ground.

STUDENT B

Pentest labs, plus builds and breaks their own infrastructure, writes, and learns one adjacent domain.

Finance, healthcare, OT, or law. Context is the expensive part, and it stays un-automatable longer.

Specialize eventually. The horizontal bar of the T comes first.

Anti-silo

This semester (technical)

01 Run a local model with Ollama. One afternoon.

02 Write one page explaining a bug to a non-technical reader.

03 Stand up one homelab service, then attack it yourself.

04 Pick one adjacent domain. Read an hour a week.

05 Send one real open-source pull request, however small.

06 Find one AI output that's wrong. Document why.

This semester (networking)

01 Find a passion, dive deeply into it. Find a way to *share your passion* - speaking gig, blog, etc.

03 Help first, ask never. *Share knowledge freely*, make introductions, and answer questions publicly.

05 *Have a take*. People remember you for opinions, not agreement. Be willing to defend your position.

02 Find *your people* - group, conference, online chat, whatever. See them regularly.

04 Stay in touch without an agenda. *Reach out* when you don't need anything.

06 Be *genuinely curious*. Ask specific questions. The best networkers are interested, not just interesting.

THREE THINGS TO REMEMBER

Machines run campaigns now.

Verification is the durable skill.

Breadth before depth.

Questions without clean answers

If an AI finds a zero-day in your company's product before you do, who's liable?

Should model weights be export-controlled?

If a proof is only checkable by machine, what does "understanding" mean?

Is a security career that depends on human scarcity a good bet?

What should this course be teaching that it isn't?

I don't have answers to these. That's the point.

Clubs, Groups, and Cons

A curated list of some of the groups around town I'd look at joining asap.

RESOURCE LINKS

DEF CON 402: dc402.org

Kernelcon: kernelcon.org

NEbraskaCERT: nebraskacert.org

OWASP: <incoming/>

Cyber Jutsu: meetup.com/omaha-women-in-cyber

ISACA: engage.isaca.org/omahachapter/home

ISC2: isc2omaha.org

Code-specific groups: omahamtg.com
(dotnet), ojug.org (java), and more!

TechOmaha:
techomaha.com/list-omaha-user-groups

Contact and resources

Feel free to connect and reach out, I'm almost always willing to talk or help.



@clevernyyyy



@appsec.bsky.social



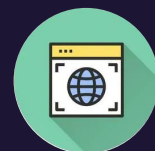
/in/adamschaal



/clevernyyyy



clevernyyyy



adamschaal.com

RESOURCE LINKS

email: `me@adamschaal.com`

slides: `adamschaal.com/s/dont-pick-a-lane.pdf`

resume-from-code: `https://offsec.sh`

ctf: `pwn.college`, `picoctf.org`

homelab: `r/homelab`, `r/selfhosted`

reading list: Security Engineering by Ross Anderson (book), CyberSecEval by Meta Puple Llama (paper)

open-source projects: OpenRewrite, Garak, PyRit